

ASSESSMENT - 7

Functional Dependencies, Normalization and Key Constraints in Relational Databases

Objective

To analyze relational databases using functional dependencies, candidate keys, normalization techniques from 1NF to 5NF, and lossless-join decomposition in order to identify redundancy and maintain data integrity.

1. Identify Functional Dependencies and Candidate Keys

Given Relation

STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade)

Where:

- SID = Student ID
- SName = Student Name
- CID = Course ID
- CName = Course Name
- Instructor = Course Instructor
- Grade = Student's Grade

Step 1: Identify Functional Dependencies

A functional dependency represents a relationship where one attribute uniquely determines another attribute.

The functional dependencies are:

1. **SID $\rightarrow$ SName**
2. **CID $\rightarrow$ CName**
3. **CID $\rightarrow$ Instructor**
4. **(SID, CID) $\rightarrow$ Grade**

Therefore,

$F = \{ \text{SID} \rightarrow \text{SName}, \text{CID} \rightarrow \text{CName}, \text{CID} \rightarrow \text{Instructor}, \text{SID}, \text{CID} \rightarrow \text{Grade} \}$

Step 2: Find Candidate Key

Consider the attribute set:

$(\text{SID}, \text{CID})^+$

Initially:

$(\text{SID}, \text{CID})^+ = \{\text{SID}, \text{CID}\}$

Using **$\text{SID} \rightarrow \text{SName}$** :

$\rightarrow \text{SName}$

Using **$\text{CID} \rightarrow \text{CName}, \text{Instructor}$** :

$\rightarrow \text{CName}, \text{Instructor}$

Using **$\text{SID}, \text{CID} \rightarrow \text{Grade}$** :

$\rightarrow \text{Grade}$

Therefore,

$(\text{SID}, \text{CID})^+ = \{\text{SID}, \text{SName}, \text{CID}, \text{CName}, \text{Instructor}, \text{Grade}\}$

Since the closure contains **all attributes** of the relation,

(SID, CID) is a superkey.

Neither SID nor CID alone can determine all attributes. Hence:

Candidate Key = (SID, CID)

Conclusion

The relation has the functional dependencies:

- $\text{SID} \rightarrow \text{SName}$
- $\text{CID} \rightarrow \text{CName}, \text{Instructor}$
- $\text{SID}, \text{CID} \rightarrow \text{Grade}$

The candidate key is **(SID, CID)** .

2. Conversion of an Unnormalized Relation into 1NF

Introduction

An unnormalized relation may contain repeating groups or multiple values in a single cell. First Normal Form (1NF) requires every attribute to contain only atomic values.

Unnormalized Relation

Consider the following STUDENT relation:

SID	SName	Courses
101	Arun	DBMS, AI, CN
102	Priya	Python, DBMS

The Courses attribute contains multiple values.

Therefore, this relation is not in 1NF.

Step 1: Identify the Repeating Group

For student 101:

Courses = DBMS, AI, CN

These are multiple values stored in one attribute.

For student 102:

Courses = Python, DBMS

Again, multiple values are present.

Step 2: Remove Repeating Groups

Each course is placed in a separate row.

1NF Relation

SID	SName	Course
101	Arun	DBMS
101	Arun	AI

SID	SName	Course
101	Arun	CN
102	Priya	Python
102	Priya	DBMS

Now each cell contains a single atomic value.

Step 3: Verify 1NF

The relation satisfies:

1. Every column contains a single type of value.
2. Every cell contains only one value.
3. There are no repeating groups.
4. Each row represents a single record.

Therefore,

The relation is in 1NF.

Conclusion

The conversion from UNF to 1NF is achieved by removing repeating groups and ensuring that every attribute contains atomic values.

3. Decomposition of R(A,B,C,D,E) into 2NF

Given

Relation:

R(A,B,C,D,E)

Functional Dependencies:

- $A \rightarrow BC$
- $BC \rightarrow D$
- $D \rightarrow E$

Step 1: Find Candidate Key

Calculate the closure of A.

A⁺

Initially:

A⁺ = {A}

Using:

A → BC

Therefore:

A⁺ = {A,B,C}

Using:

BC → D

Therefore:

A⁺ = {A,B,C,D}

Using:

D → E

Therefore:

A⁺ = {A,B,C,D,E}

Thus,

A⁺ = {A,B,C,D,E}

Since A determines all attributes:

Candidate Key = A

Step 2: Check for Partial Dependency

A relation is in 2NF if:

- It is already in 1NF.
- There is no partial dependency of a non-key attribute on a proper subset of a composite candidate key.

Here the candidate key is **A**, which contains only one attribute.

Therefore, there cannot be a proper subset of the key.

Hence, partial dependency does not exist.

Step 3: Check 2NF

Since there is no partial dependency:

R is already in 2NF.

No decomposition is required.

Conclusion

The relation **R(A,B,C,D,E)** is in **2NF** because its candidate key is a single attribute **A**, so partial dependency is impossible.

Note: The dependency $D \rightarrow E$ is a transitive dependency, so the relation is not necessarily in 3NF. Further decomposition would be needed for 3NF.

4. 3NF Decomposition of EMPLOYEE Relation

Given Relation

EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID)

Functional Dependencies

1. $\text{EmpID} \rightarrow \text{EmpName}, \text{DeptID}$
2. $\text{DeptID} \rightarrow \text{DeptName}, \text{ManagerID}$

Step 1: Determine Candidate Key

EmpID determines:

$\text{EmpID} \rightarrow \text{EmpName}, \text{DeptID}$

DeptID determines:

$\text{DeptID} \rightarrow \text{DeptName}, \text{ManagerID}$

Therefore:

$\text{EmpID} \rightarrow \text{EmpName}, \text{DeptID}, \text{DeptName}, \text{ManagerID}$

Hence,

Candidate Key = EmpID

Step 2: Identify Transitive Dependency

We have:

$\text{EmpID} \rightarrow \text{DeptID}$

and

$\text{DeptID} \rightarrow \text{DeptName}, \text{ManagerID}$

Therefore:

$\text{EmpID} \rightarrow \text{DeptName}, \text{ManagerID}$

This is a transitive dependency.

The dependency can be represented as:

$\text{EmpID} \rightarrow \text{DeptID} \rightarrow \text{DeptName}, \text{ManagerID}$

This violates **3NF**.

Step 3: Decompose the Relation

EMPLOYEE

EMPLOYEE(EmpID, EmpName, DeptID)

EmpID	EmpName	DeptID
E101	Arun	D01
E102	Priya	D02

DEPARTMENT

DEPARTMENT(DeptID, DeptName, ManagerID)

DeptID	DeptName	ManagerID
D01	CSE	M01
D02	AI&DS	M02

Step 4: Verify 3NF

In EMPLOYEE:

EmpID $\rightarrow$ EmpName, DeptID

EmpID is the primary key.

In DEPARTMENT:

DeptID $\rightarrow$ DeptName, ManagerID

DeptID is the primary key.

Therefore, transitive dependency has been removed.

Conclusion

The original relation has been successfully decomposed into:

- EMPLOYEE(EmpID, EmpName, DeptID)
- DEPARTMENT(DeptID, DeptName, ManagerID)

Both relations satisfy **3NF**.

5. BCNF Analysis and Decomposition

Given Relation

R(A,B,C,D)

Functional Dependencies:

- $AB \rightarrow C$
- $C \rightarrow D$
- $D \rightarrow A$

Step 1: Find Candidate Keys

Calculate AB^+ .

Initially:

$AB^+ = \{A,B\}$

Using:

$AB \rightarrow C$

We get:

$AB^+ = \{A,B,C\}$

Using:

$C \rightarrow D$

We get:

$AB^+ = \{A,B,C,D\}$

Therefore, AB is a candidate key.

Now calculate BC^+ :

B,C

Using:

$C \rightarrow D$

$\rightarrow D$

Using:

$D \rightarrow A$

$\rightarrow A$

Therefore:

$BC^+ = \{A, B, C, D\}$

So BC is also a candidate key.

Similarly, **CD** is a candidate key because:

$C \rightarrow D$

$D \rightarrow A$

and with C,D we obtain A, but B cannot be obtained. So CD is **not** a key.

Thus candidate keys include:

AB and BC

Step 2: BCNF Condition

A relation is in BCNF if, for every non-trivial FD:

$X \rightarrow Y$

X must be a **superkey**.

Check:

$AB \rightarrow C$

AB is a candidate key.

Therefore, it satisfies BCNF.

$C \rightarrow D$

C is not a superkey.

Therefore, **BCNF is violated**.

$D \rightarrow A$

D is not a superkey.

Therefore, **BCNF is also violated**.

Step 3: Decompose Using $C \rightarrow D$

Create:

R1(C,D)

and

R2(A,B,C)

Step 4: Verify

For R1:

$C \rightarrow D$

C is the key of R1.

Therefore, R1 is in BCNF.

For R2:

$AB \rightarrow C$

AB is a key of R2.

Therefore, R2 satisfies BCNF.

Final Decomposition

R1(C,D)

R2(A,B,C)

Conclusion

The original relation was not in BCNF because $C \rightarrow D$ and $D \rightarrow A$ have determinants that are not superkeys. It is decomposed into BCNF relations **R1(C,D)** and **R2(A,B,C)**.

6. Lossless-Join Decomposition Using Chase Algorithm

Consider:

R(A,B,C)

Decomposition:

- R1(A,B)
- R2(B,C)

Assume the functional dependency:

B → C

Step 1: Construct Chase Table

Relation A B C

R1(A,B) a b c₁

R2(B,C) a₂ b c

The common attribute between R1 and R2 is **B**.

Step 2: Apply Functional Dependency

Given:

B → C

Both rows have the same value **b** for B.

Therefore, the value of C must become the same.

Thus, C becomes the common value.

Relation A B C

R1 a b c

R2 a₂ b c

Step 3: Check Losslessness

A decomposition is lossless if the chase produces a row containing the distinguished symbols for all attributes.

The common attributes allow the original information to be reconstructed.

Therefore:

The decomposition is lossless.

Conclusion

The Chase algorithm verifies that the decomposition does not introduce spurious tuples and preserves the information of the original relation.

7. Multivalued Dependencies and 4NF

Given Relation

TEACHER(TID, Subject, Hobby)

Consider:

TID	Subject	Hobby
T1	DBMS	Music
T1	DBMS	Drawing
T1	AI	Music
T1	AI	Drawing

A teacher can teach multiple subjects and can have multiple hobbies.

The subjects and hobbies are independent of each other.

Step 1: Identify Multivalued Dependencies

The multivalued dependencies are:

TID $\twoheadrightarrow$ Subject

and

TID $\twoheadrightarrow$ Hobby

This means a TID can determine multiple independent values of Subject and Hobby.

Step 2: Identify Redundancy

For teacher T1:

Subjects:

- DBMS
- AI

Hobbies:

- Music

- Drawing

All possible combinations are stored.

This causes unnecessary repetition.

Step 3: Decompose into 4NF

Create two relations.

TEACHER_SUBJECT

TEACHER_SUBJECT(TID, Subject)

TID	Subject
T1	DBMS
T1	AI

TEACHER_HOBBY

TEACHER_HOBBY(TID, Hobby)

TID	Hobby
T1	Music
T1	Drawing

Conclusion

The multivalued dependencies have been separated into independent relations. Hence, the decomposition eliminates redundancy and satisfies **4NF**.

8. Join Dependencies and 5NF

Introduction

A **join dependency** occurs when a relation can be reconstructed by joining multiple smaller relations.

Consider:

R(Employee, Project, Skill)

Suppose employees work on projects and possess different skills.

Storing all three attributes together can produce redundant combinations.

Example

Employee	Project	Skill
E1	P1	Java
E1	P1	Python
E1	P2	Java
E1	P2	Python

If employee E1 works on P1 and P2 and possesses Java and Python skills, several combinations are stored.

This creates redundancy.

Step 1: Decompose

Create:

EMPLOYEE_PROJECT

EP(Employee, Project)

EMPLOYEE_SKILL

ES(Employee, Skill)

PROJECT_SKILL

PS(Project, Skill)

Step 2: Join the Relations

The original relation can be reconstructed by joining the decomposed relations according to the applicable join dependency.

The decomposition removes redundancy caused by storing independent relationships together.

Conclusion

A relation is in **5NF/PJNF** when every non-trivial join dependency is implied by the candidate keys.

The decomposition into smaller relations eliminates join-related redundancy and improves data integrity.

9. Anomalies and Normalization of College Database

Consider the poorly designed relation:

COLLEGE(SID, SName, DeptID, DeptName, HOD)

Example

SID	SName	DeptID	DeptName	HOD
101	Arun	D01	CSE	Dr. Kumar
102	Priya	D01	CSE	Dr. Kumar
103	Ravi	D02	AI&DS	Dr. Meena

A. Insertion Anomaly

Suppose a new department D03 is created, but no student has joined it.

We cannot insert department details without providing a SID.

Therefore, department information cannot be independently inserted.

B. Update Anomaly

Suppose the HOD of D01 changes from Dr. Kumar to Dr. Raj.

The HOD value must be changed in every row belonging to D01.

If one row is not updated, inconsistent data occurs.

C. Deletion Anomaly

Suppose student 103 is the only student in D02.

If student 103 is deleted, the information about D02 and its HOD is also lost.

Step 1: Functional Dependencies

SID $\rightarrow$ SName, DeptID

DeptID $\rightarrow$ DeptName, HOD

Step 2: Identify Transitive Dependency

We have:

SID → DeptID

and

DeptID → DeptName, HOD

Therefore:

SID → DeptName, HOD

This is a transitive dependency.

Step 3: Decompose into 3NF

STUDENT

STUDENT(SID, SName, DeptID)

SID	SName	DeptID
101	Arun	D01
102	Priya	D01
103	Ravi	D02

DEPARTMENT

DEPARTMENT(DeptID, DeptName, HOD)

DeptID	DeptName	HOD
D01	CSE	Dr. Kumar
D02	AI&DS	Dr. Meena

Conclusion

The original college database contains:

- Insertion anomaly
- Update anomaly
- Deletion anomaly

After decomposition into STUDENT and DEPARTMENT, redundancy is reduced and the database satisfies **3NF**.

10. Attribute Closure and Candidate Keys

Consider the relation:

R(A,B,C,D)

Functional Dependencies:

- $A \rightarrow B$
- $B \rightarrow C$
- $C \rightarrow D$

Step 1: Calculate A^+

Start with:

$$A^+ = \{A\}$$

Using $A \rightarrow B$:

$$A^+ = \{A, B\}$$

Using $B \rightarrow C$:

$$A^+ = \{A, B, C\}$$

Using $C \rightarrow D$:

$$A^+ = \{A, B, C, D\}$$

Therefore:

$$A^+ = \{A, B, C, D\}$$

Since A determines all attributes:

A is a Candidate Key.

Step 2: Calculate B^+

Start:

$$B^+ = \{B\}$$

Using $B \rightarrow C$:

$$B^+ = \{B, C\}$$

Using $C \rightarrow D$:

$$B^+ = \{B, C, D\}$$

A cannot be obtained.

Therefore:

B is not a candidate key.

Step 3: Calculate C^+

Start:

$$C^+ = \{C\}$$

Using $C \rightarrow D$:

$$C^+ = \{C, D\}$$

A and B cannot be obtained.

Therefore:

C is not a candidate key.

Step 4: Calculate D^+

There is no FD beginning with D.

Therefore:

$$D^+ = \{D\}$$

D cannot determine the remaining attributes.

Therefore:

D is not a candidate key.

Closure Table

Attribute Set	Closure	Candidate Key?
A	$\{A, B, C, D\}$	Yes
B	$\{B, C, D\}$	No
C	$\{C, D\}$	No

Attribute Set	Closure	Candidate Key?
D	{D}	No

Final Answer

Candidate Key = A

Overall Conclusion

Functional dependencies and normalization are essential for designing an efficient relational database. **1NF** removes repeating groups, **2NF** eliminates partial dependencies, **3NF** removes transitive dependencies, **BCNF** provides a stronger form of dependency-based normalization, **4NF** eliminates unwanted multivalued dependencies, and **5NF** handles join dependencies. These normalization techniques reduce redundancy, prevent insertion/deletion/update anomalies, and maintain data consistency and integrity.